Antonio Maria Sudoso

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Research Interests

Optimization, Machine Learning, Time Series.

Education and Current Position

Postdoctoral Researcher
Sapienza University of Rome, Italy — *Feb 2022 –*
Department of Computer, Control and Management Engineering "A. Ruberti"

PhD degree in Computer Science Control and Geoinformation
University of Rome Tor Vergata, Italy, Laude grade — *May 2022*
Thesis: "Global Optimization for Machine Learning Applications"
Advisor: Veronica Piccialli

Visiting PhD Student
Alpen-Adria-Universität Klagenfurt, Klagenfurt am Wörthersee, Austria — *Jan 2020*
Hosted by: Angelika Wiegele

Laurea Magistrale (MSc equivalent) in Computer Engineerinig
University of Rome Tor Vergata, Italy, 110/110 cum laude — *Oct 2018*
Thesis: "Deep Neural Networks for Time Series Analysis"
Advisors: Paolo Mancuso and Veronica Piccialli

Laurea (BSc equivalent) in Computer Engineerinig
University of Rome Tor Vergata, Italy, 110/110 — *Oct 2016*

Work Experience

Deix S.r.l.
Data Scientist — *July 2021–Dec 2021*
I developed optimization algorithms for point cloud registration and segmentation.

Wsense S.r.l.
Data Scientist — *June 2021–Dec 2021*
I developed algorithms for anomaly detection of fish behavior in aquaculture monitoring systems.

Professional Activities

Reviewer for Journals

I reviewed manuscripts submitted to the following international journals:

- International Journal of Forecasting, Elsevier (2022)
- Journal of Grid Computing, Springer (2022)
- INFORMS Journal on Computing, INFORMS (2022)

Collaborations

I have been involved in collaboration agreements between ENEA and University of Rome Tor Vergata:

- C. La Riccia, V. Piccialli and A. M. Sudoso (PAR 2018, 10.000€). "Clustering di tipologie di abitazione per scegliere modelli di disaggregazione di consumi elettrici realizzati tramite reti deep".
- G. Oriolo, V. Piccialli and A. M. Sudoso (PAR 2019, 40.000€). "Disaggregazione dei consumi elettrici domestici tramite modelli di machine learning integrati con tecniche di previsione di consumo elettrico".

Research Output

Journal Papers

J4 V. Piccialli, A. M. Sudoso and A. Wiegele (2022). SOS-SDP: An Exact Solver for Minimum Sum-of-Squares Clustering, *INFORMS Journal on Computing*. `doi.org/10.1287/ijoc.2022.1166`

J3 M. Balletti, V. Piccialli and A. M. Sudoso (2022). Mixed-Integer Nonlinear Programming for State-based Non-Intrusive Load Monitoring, *IEEE Transactions on Smart Grid*, vol. 13, no. 4, pp. 3301-3314. `doi.org/10.1109/TSG.2022.3152147`

J2 P. Mancuso, V. Piccialli and A. M. Sudoso (2021). A machine learning approach for forecasting hierarchical time series, *Expert Systems with Applications*, 182, 115102. `doi.org/10.1016/j.eswa.2021.115102`

J1 V. Piccialli and A. M. Sudoso (2021). Improving Non-Intrusive Load Disaggregation through an Attention-Based Deep Neural Network, *Energies*, 14(4), 847. `doi.org/10.3390/en14040847`

Submitted Papers

S1 V. Piccialli, A. Russo Russo and A. M. Sudoso (2021). An Exact Algorithm for Semi-supervised Minimum Sum-of-Squares Clustering. `https://arxiv.org/abs/2111.15571`

Teaching

Teaching Assistant

As PhD Student, I was involved in the following teaching activities:

- Machine Learning (from 2018/2019 to 2020/2021), MSc in Computer Engineering, University of Rome Tor Vergata.
- Optimization Methods for Big Data (from 2018/2019 to 2020/2021), MSc in Computer Engineering, University of Rome Tor Vergata.

Regular Courses

- Hands-on Laboratory of Machine Learning (2018-2019), 1st level Master degree in Data Science, University of Rome Tor Vergata, 16 hours.
- Hands-on Laboratory of Machine Learning (2019-2020), 1st level Master degree in Data Science, University of Rome Tor Vergata, 16 hours.
- Hands-on Laboratory of Machine Learning (2020-2021), advanced course in Data Science for the employees of *Corte dei Conti*, University of Rome Tor Vergata, 16 hours.

- Hands-on Laboratory of Machine Learning (2020-2021), 1st level Master degree in Data Science, University of Rome Tor Vergata, 16 hours.
- PhD Course on Advances in Operations Research (ABRO 2022), Clustering and Optimization, Sapienza University of Rome, 2 hours.

Thesis Supervision

I have co-supervised 11 *Laurea Magistrale* (MSc) theses.

Talks

2021 Machine Learning NeEDS Mathematical Optimization, YOUNG Seminar Series `https://congreso.us.es/mlneedsmo/`, *SOS-SDP: an Exact Solver for Minimum Sum-of-Squares Clustering*.

2021 SIAM Conference on Optimization (OP21), SDP Approaches to Combinatorial and Global Optimization, *An SDP-based approach for minimum sum-of-square clustering*.

2021 International Conference on Optimization and Decision Sciences (ODS21), Data Science, *SOS-SDP: an Exact Solver for Minimum Sum-of-Squares Clustering*.

Awards and Grants

2021 Winner of $1000 Google Cloud Research Credits program with the funding number GCP19980904 for the research project *Global Optimization for Semi-supervised Sum-of-Squares Clustering*.

Schools and Workshops

2019 Optimization, Big Data and Applications (OBA19), Summer School, Veroli, Italy.

2020 New Advances in Optimization, Machine Learning and Data Science, 4th AIROYoung Workshop, Bozen, Italy.

2021 EUROPT Summer School 2021, Summer School, Online.